RESEARCH DEVELOPMENT / REVISED SEPTEMBER 26, 2026

Mathematical Repairs and Covariance-Aware Inference

for *Machine Markets*

Corrected identification theorems, continuous-time stability, finite-sample ellipsoids, conic outer bounds, and a convergent refinement procedure.

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Manuscript basis: Formulated initially against `Machine_Markets_Rebuilt.pdf` (56 pages, September 24, 2026); updated and integrated with the revised 8-chapter master manuscript (`Machine_Markets_Final_Integrated_Paper.pdf`) and verified against the repository implementation (`research_extension/mm_revision`).

Status: Integrated mathematical supplement providing analytical proofs, covariance-aware conic derivations, and experimental protocols. It is coordinated with the master manuscript and repository test suite.

Central correction. The feedback quantity $S = a + Q$ is generally not a fixed linear contrast of finitely many public impulse-response coefficients. Covariance-aware inference must retain the response-to-model compatibility conditions.

Main construction. A finite-sample Gaussian response ellipsoid can be combined with positive-memory constraints through a second-order cone outer relaxation. The relaxation preserves continuous rate support, admits rigorous endpoint bounds, and can be refined toward the interval hull of a declared compact model-compatible set.

Evidence boundary. Written proofs cannot supply missing market records, establish actual runtime provenance, measure a comparator's performance, or demonstrate future forecasting gains. Those remain separate empirical and implementation tasks.

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S1 What this supplement resolves, and why each part matters

The existing manuscript supplies a positive-feedback model, finite-window witnesses, a Gaussian episode measurement contract, a rectangular recurrence relaxation, audited terminal-row inference, and a funded-system counterexample [1]. The extensions below are explicitly separate from those retained results.

Gap	Resolution developed here	Scientific reason and remaining boundary
Continuous time	Diagonal-rate continuous-time stability theorem; correctly scaled Euler limit; exact-sampling distinction.	Removes a time-unit ambiguity. It does not make the market model automatically realistic or establish a new general positive-systems theorem.
Group attribution	Known-rate rank theorem; unknown-rate identified set; known-total normalization; four-tap aggregate recovery.	Prevents an identifiability claim from silently changing its available information.
Response covariance	Exact F-calibrated ellipsoid under the manuscript's independent Gaussian episode law.	Uses full covariance without treating its estimate as the true covariance.
Structural gain	Model-compatible optimization rather than the incorrect formula $S = c^\top \beta$.	Keeps finite-window non-identification and uncertain hidden memory in the inference problem.
Convex computation	Continuous-rate conic outer relaxation, endpoint-validity proof, and bounded-domain dual certificates.	Exact or outward-valid bounds are required; the numerical implementation presently withholds certification.
Sharpness	Exhaustive response-box and rate-bin refinement with a compactness convergence proof.	Shows a route to arbitrary sharpness for the stated compact class, without promising efficient convergence.
Decision errors	Coverage-to-action theorem and a two-world abstention bound.	Separates unconditional errors from errors conditional on issued labels.
Economic scope	Formal non-transfer example for solvency; empirical and verification checklist.	Prevents a signal theorem from being advertised as a financial safety certificate.

Priority and originality. Positive-system stability, Hotelling regions, ellipsoid support functions, McCormick inequalities, conic duality, and compactness arguments are established tools [2, 3, 5, 6]. Their application and assembly here need a literature-priority review before being described as original. The specific conic construction below provides the formal theoretical derivation for the second-order cone outer relaxation implemented via HiGHS cutting planes with dual residual repair (`research_extension.mm_revision.inference.conic_certificate.fit_conic_certificate`), with primary empirical benchmarks executed in Chapter 4 of the main manuscript.

Scope of completeness. Every formal statement below is proved for its stated mathematical setting. No proof assistant has checked this supplement. No claim is made that all possible market-identification problems, all empirical gaps, or all versions of the covariance-comparison question are resolved.

S2 Canonical notation and observation contracts

In the retained discrete pressure-coordinate model,

$$x_{t+1} = ax_t + \sum_{i=1}^{n} y_{i,t} + \kappa u_t + e_{t+1}, \tag{S1}$$

$$y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i q_i x_t, \tag{S2}$$

where $0 \le a < 1$, $q_i \ge 0$, $0 < \eta_i \le 1$, and $\kappa > 0$. Set

$$Q = \sum_i q_i, \qquad S = a + Q, \qquad G = \frac{Q}{1-a}.$$

These are different quantities. In this model, $S < 1$ is equivalent to Schur stability, but S is not the numerical spectral radius. Nor is G the closed-loop steady-state multiplier: for a stable system and a constant input, that multiplier is $\kappa/(1-a-Q)$.

With zero adaptive initial state, define

$$k_j = \sum_i q_i \eta_i (1-\eta_i)^j, \tag{S3}$$

$$h_0 = 1, \quad h_1 = a, \quad h_{t+1} = ah_t + \sum_{j=0}^{t-1} k_j h_{t-1-j}, \tag{S4}$$

$$\beta_j = \kappa h_j. \tag{S5}$$

The observable response is β, not the private state vector and not the parameter vector (a, q).

I0: controlled price responses. The observer receives randomized probe magnitudes and episode price responses, together with an externally justified rate floor. Exact rates, private states, parameters, and covariance are not supplied.

I1: audited architecture. Exact prices, disclosed rates, verified resets, and a complete relevant channel inventory permit reconstructed regressors and the terminal-row pivot. This is stronger information than I0.

I2: observational exchange records. Public quotes and trades do not automatically provide exogenous interventions, independent repetitions, exact resets, or the maintained structural model. The theorems below apply to I2 only after the relevant observation and sampling assumptions are established.

S3 Continuous-time stability and the correct sampling limit

S3.1 A positive-vector lemma

A real matrix is *Metzler* when every off-diagonal entry is nonnegative. It is *Hurwitz* when all eigenvalues have negative real part. The following established criterion is included with a proof so the application is self-contained [2].

Lemma S3.1 (Positive-vector characterization)**.** *For a finite-dimensional Metzler matrix A,*

$$A \text{ is Hurwitz} \iff \exists v > 0 \text{ such that } Av < 0.$$

All inequalities between vectors are componentwise.

Proof. Choose c large enough that $A + cI \geq 0$. Then $e^{At} = e^{-ct}e^{(A+cI)t} \geq 0$ for $t \geq 0$, by the exponential series. If A is Hurwitz, define

$$v = \int_0^\infty e^{At}\mathbf{1}\, dt.$$

The integral converges, and every coordinate is positive because the integrand equals $\mathbf{1}$ at zero and is continuous. Integrating its derivative yields $Av = -\mathbf{1} < 0$.

Conversely, if $v > 0$ and $Av < 0$, let $D = \mathrm{diag}(v)$ and $\widetilde{A} = D^{-1}AD$. This matrix is Metzler, similar to A, and has strictly negative row sums. The rightmost real coordinate of its ith Gershgorin disk is its row sum. Every disk therefore lies in the open left half-plane, so every eigenvalue of A has negative real part. □

Lemma S3.2 (Nonnegative discrete counterpart)**.** *For $P \geq 0$, $\rho(P) < 1$ if and only if some $v > 0$ satisfies $Pv < v$.*

Proof. If $\rho(P) < 1$, the Neumann series gives $v = (I-P)^{-1}\mathbf{1} > 0$ and $Pv = v - \mathbf{1} < v$. Conversely, scaling by $D = \mathrm{diag}(v)$ gives a nonnegative matrix with all row sums strictly below one. Its induced infinity norm, hence its spectral radius, is below one. □

S3.2 Heterogeneous continuous rates

Let

$$\Gamma = \mathrm{diag}(\gamma_1, \ldots, \gamma_d) > 0, \qquad E = \mathrm{diag}(\lambda_1, \ldots, \lambda_n) > 0,$$

and let $B \in \mathbb{R}_+^{d\times n}$, $C \in \mathbb{R}_+^{n\times d}$, and $g \geq 0$. Consider

$$\dot{x} = -\Gamma x + gBm, \qquad \dot{m} = E(Cx - m), \qquad A_c = \begin{bmatrix} -\Gamma & gB \\ EC & -E \end{bmatrix}. \tag{S6}$$

Theorem S3.3 (Continuous-time positive-feedback boundary)**.** *The system (S6) is exponentially asymptotically stable if and only if*

$$\rho\Big(\Gamma^{-1}gBC\Big) < 1. \tag{S7}$$

In particular, for $\Gamma = \gamma I_d$,

$$A_c \text{ is Hurwitz} \iff g\rho(BC) < \gamma.$$

This condition is independent of the positive diagonal memory rates in E.

Proof. Let $K = \Gamma^{-1}gBC \geq 0$. Suppose $\rho(K) < 1$ and put $x = (I - K)^{-1}\mathbf{1} > 0$. Then

$$(\Gamma - gBC)x = \Gamma\mathbf{1} > 0.$$

For sufficiently small $\varepsilon > 0$, set $m = Cx + \varepsilon\mathbf{1} > 0$. The two block components of $A_c(x, m)^\top$ are

$$-\Gamma x + gBm = -\Gamma\mathbf{1} + \varepsilon gB\mathbf{1} < 0, \qquad ECx - Em = -\varepsilon E\mathbf{1} < 0.$$

Lemma S3.1 proves that A_c is Hurwitz.

Conversely, if A_c is Hurwitz, the same lemma provides $x > 0, m > 0$ such that $-\Gamma x + gBm < 0$ and $ECx - Em < 0$. Because E is positive diagonal, $m > Cx$. Nonnegativity then gives

$$\Gamma x > gBm \geq gBCx, \qquad Kx < x.$$

Lemma S3.2 implies $\rho(K) < 1$. □

Proposition S3.4 (Boundary and supercritical behavior)**.** *If $\rho(\Gamma^{-1}gBC) = 1$, A_c has a zero eigenvalue. If that spectral radius exceeds one, A_c has a strictly positive real eigenvalue.*

Proof. At equality, Perron–Frobenius gives $x \geq 0$, $x \neq 0$, with $gBCx = \Gamma x$. Set $m = Cx$. Then $A_c(x, m)^\top = 0$.

For the strict case define, for $s \geq 0$,

$$K(s) = g(sI + \Gamma)^{-1}B(sI + E)^{-1}EC.$$

Its spectral radius is continuous, exceeds one at $s = 0$, and tends to zero as $s \to \infty$. Some $s_* > 0$ therefore satisfies $\rho(K(s_*)) = 1$. Choose a corresponding nonzero nonnegative eigenvector x and set $m = (s_*I + E)^{-1}ECx$. Substitution gives $A_c(x, m)^\top = s_*(x, m)^\top$. □

Equality does not imply exponential divergence: it has at least a nondecaying mode. Supercriticality supplies an exponentially growing mode, not a claim that every initial condition grows or that a financial price must crash. Linear ODE solutions remain finite at every finite time. The result depends on positivity and fixed finite-dimensional coefficients; signed feedback and switching require other arguments.

S3.3 Common-rate spectral abscissa

Proposition S3.5 (Closed expression in the common-rate case)**.** *For $\Gamma = \gamma I_d$, $E = \lambda I_n$, and $\nu = \rho(BC)$,*

$$\mathrm{sa}(A_c) = \frac{-(\gamma + \lambda) + \sqrt{(\gamma - \lambda)^2 + 4g\lambda\nu}}{2}. \tag{S8}$$

Here $\mathrm{sa}(A_c) = \max\{\Re s : s \in \sigma(A_c)\}$.

Proof. Let the right side be s_+. If $g\nu > 0$, select $v \geq 0$ with $BCv = \nu v$. Then $Cv \neq 0$ and $w = \lambda Cv/(s_+ + \lambda)$ produces an eigenvector (v, w) at eigenvalue s_+. For any eigenvalue s other than $-\gamma, -\lambda$, block elimination gives an eigenvalue

$$\frac{(s + \gamma)(s + \lambda)}{g\lambda}$$

of BC. If $\Re s > s_+$, then $\Re s > \max(-\gamma, -\lambda)$ and

$$|(s + \gamma)(s + \lambda)| \geq (\Re s + \gamma)(\Re s + \lambda) > (s_+ + \gamma)(s_+ + \lambda) = g\lambda\nu,$$

a contradiction. The excluded eigenvalues have real part at most s_+. When $g = 0$, block triangularity proves the formula. When $\nu = 0$ and $g > 0$, the same elimination restricts eigenvalues to $-\gamma, -\lambda$; both occur in the resulting characteristic polynomial, giving $s_+ = -\min(\gamma, \lambda)$. □

The proof does not apply a real-coefficient quadratic test to potentially complex eigenvalues of BC.

S3.4 Euler scaling, convergence, and step-size restrictions

For a physical step $h > 0$, forward Euler gives $P_h = I + hA_c$. In the common-rate notation of the original paper this means

$$a_h = 1 - h\gamma, \qquad g_h = hg, \qquad \eta_h = h\lambda. \tag{S9}$$

One must scale the coupling and memory-update parameters, not only rename $1 - a$.

Theorem S3.6 (Euler-consistent boundary and convergence)**.** *If*

$$0 < h \le \min\left\{\min_i \gamma_i^{-1}, \min_j \lambda_j^{-1}\right\},$$

then $P_h \ge 0$ *and*

$$P_h \text{ is Schur} \quad \Longleftrightarrow \quad A_c \text{ is Hurwitz.}$$

For common γ, λ*, this is exactly* $g_h \rho(BC) < 1 - a_h$*. In any induced matrix norm, for* $kh \le T$*,*

$$\left\|(I + hA_c)^k - e^{khA_c}\right\| \le \frac{1}{2} Th \left\|A_c\right\|^2 e^{T\|A_c\|}. \tag{S10}$$

Proof. The step-size restriction makes every entry of P_h nonnegative. If $A_c v < 0$ for some $v > 0$, then $P_h v < v$, so P_h is Schur. Conversely, if P_h is Schur, Lemma S3.2 gives $P_h v < v$, hence $A_c v < 0$. Lemma S3.1 completes the stability equivalence. Substitution of (S9) gives the common-rate inequality.

The exponential series bounds the one-step defect by $\left\|e^{hA_c} - I - hA_c\right\| \le \frac{1}{2} h^2 \left\|A_c\right\|^2 e^{h\|A_c\|}$. Use the telescoping identity for $P_h^k - (e^{hA_c})^k$, together with $\left\|P_h^j\right\| \le e^{jh\|A_c\|}$. There are k terms, each bounded by $\frac{1}{2} h^2 \left\|A_c\right\|^2 e^{kh\|A_c\|}$. Since $kh \le T$, (S10) follows. It gives convergence at grid points; piecewise-linear interpolation adds an $O(h)$ term on every bounded time interval. □

Remark S3.7 (Exact sampling is different)*.* The exact sampled transition is e^{hA_c}. Its eigenvalues are e^{hs_i}, so it is Schur exactly when A_c is Hurwitz, for every $h > 0$. It need not retain the original simple Euler block structure. Forward Euler outside its admissible step range can be unstable even for a stable ODE: $A_c = -I$, $h = 3$ gives $I + hA_c = -2I$. Thus the correct claim is not unrestricted discretization invariance.

S3.5 Continuous hidden-memory tails

For $\dot{y}_i = -\lambda_i y_i + \lambda_i q_i x$, zero initial memory gives

$$y_i(t) = q_i \lambda_i \int_0^t e^{-\lambda_i (t-s)} x(s)\, ds.$$

Hence the loop-open kernel is $K(t) = \sum_i q_i \lambda_i e^{-\lambda_i t}$ and

$$\int_0^\infty K(t)\, dt = Q, \qquad \int_T^\infty K(t)\, dt = \sum_i q_i e^{-\lambda_i T} \le Q_{\max} e^{-\lambda_0 T}$$

when $Q \le Q_{\max}$ and every $\lambda_i \ge \lambda_0 > 0$. These identities follow by direct integration. They extend the audited-tail interpretation, not the ability to verify a rate floor from prices.

S4 Corrected group identification, including unknown rates

S4.1 Aggregate equivalence

Under a common rate η, summing (S2) gives

$$Y_{t+1} = (1 - \eta) Y_t + \eta Q x_t, \qquad Y_t = \sum_i y_{i,t}.$$

For the same initial aggregate state, probe sequence, and aggregate disturbance law, prices depend on q only through Q. Equal Q therefore implies equal aggregate price laws. This is the retained symmetry result, not a statement about known heterogeneous rates.

S4.2 Known-rate measurement theorem

Let experiment j inject the channel-input vector $U_{j,:}$ at time zero, with known pulse units and otherwise zero inputs. From a zero state,

$$y_{i,1} = \eta q_i U_{j,i}, \qquad x_1 = 0, \qquad x_2 = \eta U_{j,:} q.$$

Stacking the measured lag-2 responses yields $b = \eta U q$.

Theorem S4.1 (Uniform identification with a known common rate)**.** *Fix known $\eta > 0$. The map $q \mapsto \eta U q$ is injective over all $q \geq 0$ if and only if* $\operatorname{rank} U = n$. *If a known total $Q > 0$ is included and the class is $\{q \geq 0 : \mathbf{1}^\top q = Q\}$, uniform injectivity holds if and only if*

$$\operatorname{rank} \begin{bmatrix} \mathbf{1}^\top \\ U \end{bmatrix} = n.$$

Thus $n - 1$ appropriately independent channel measurements can suffice when Q is known. Pointwise uniqueness at a boundary point is weaker than uniform injectivity.

Proof. Full column rank gives a unique solution of the linear system; in the first case it is $q = \eta^{-1}(U^\top U)^{-1} U^\top b$. If U is rank deficient, choose $z \neq 0$ in its nullspace and a strictly positive q_0. For sufficiently small $\varepsilon > 0$, both $q_0 \pm \varepsilon z$ are nonnegative and yield the same response. This proves necessity for uniform identification.

With a fixed positive Q, use the augmented matrix. If it is rank deficient, its nullspace contains $z \neq 0$ with $Uz = 0$ and $\mathbf{1}^\top z = 0$. Perturb the interior point $q_0 = (Q/n)\mathbf{1}$ as above. Both alternatives preserve the sum and observations. Conversely, full column rank of the augmented matrix gives uniqueness. Taking U to measure the first $n - 1$ coordinates realizes the sufficient count. For $Q = 0$, nonnegativity already forces $q = 0$, regardless of rank; this is why the necessity statement excludes that degenerate simplex. □

S4.3 Unknown rate: an exact identified-set description

Assume only $0 < \eta_{\min} \leq \eta \leq \eta_{\max} \leq 1$ and observe b, U. Set $s = 1/\eta$.

Theorem S4.2 (Unknown-rate set and normalization)**.** *The exact group-parameter identified set from lag-2 channel observations is the projection onto q of the polyhedron*

$$\mathcal{P}_b = \left\{ (q, s) : q \geq 0, \quad Uq = sb, \quad \eta_{\max}^{-1} \leq s \leq \eta_{\min}^{-1} \right\}. \tag{S11}$$

A known total is incorporated by the additional equality $\mathbf{1}^\top q = Q$. If U has full column rank and $v = U^\dagger b$ is nonnegative and consistent with $Uv = b$, the set without a total is $\{q = sv : s \in [\eta_{\max}^{-1}, \eta_{\min}^{-1}]\}$. If additionally $Q > 0$ and $\mathbf{1}^\top v > 0$, then, when compatible with the rate interval,

$$\eta = \frac{\mathbf{1}^\top v}{Q}, \qquad q = \frac{Qv}{\mathbf{1}^\top v}. \tag{S12}$$

Proof. The map $\eta \mapsto s = 1/\eta$ is a bijection between the declared rate interval and the interval in (S11). Multiplying $b = \eta U q$ by s gives exactly $Uq = sb$, proving both directions of the set representation. With full column rank, $Uq = sb$ implies $q = sv$. The known-total constraint then gives $s\mathbf{1}^\top v = Q$, hence (S12). If this s is outside its allowed interval, the observations and side information conflict. If $v = 0$, full rank forces $q = 0$ but does not identify η. Rank deficiency can leave non-radial nullspace ambiguity, so a universal "ray only" statement is false. □

For example, $U = I_2$, $b = (0.06, 0.02)^\top$, $Q = 0.4$ identify $\eta = 0.2$ and $q = (0.3, 0.1)^\top$. Without Q, both $(\eta, q) = (0.2, (0.3, 0.1))$ and $(0.4, (0.15, 0.05))$ fit the same lag-2 data.

The polyhedral formulation is for exact b. If b itself is uncertain inside an ellipsoid, $Uq = sb$ again contains a product; one must propagate that uncertainty rather than reuse the exact-data LP as a confidence procedure. For full-column-rank U, zero observations identify $q = 0$; the rate is unidentified only when its allowed interval is nondegenerate. If $\eta_{\min} = \eta_{\max}$, the rate is known. For a fixed known rate and feasible q_0, the identified set is $(q_0 + \ker U) \cap \mathbb{R}^n_+$, with local dimension $n - \operatorname{rank} U$ at an interior point. Boundary uniqueness does not imply uniform injectivity.

S4.4 The aggregate common rate may itself be identifiable

Proposition S4.3 (Four-tap aggregate recovery)**.** *In the common-rate model, suppose exact $\beta_0, \beta_1, \beta_2, \beta_3$ are available, $\beta_0 = \kappa > 0$, and $Q > 0$. Define $h_j = \beta_j/\beta_0$ and*

$$a = h_1, \qquad k_0 = h_2 - a^2, \qquad k_1 = h_3 - 2ah_2 + a^3.$$

Then

$$\eta = 1 - \frac{k_1}{k_0}, \qquad Q = \frac{k_0}{\eta}, \qquad \kappa = \beta_0. \tag{S13}$$

Proof. The first recurrences give $h_2 = a^2 + k_0$ and $h_3 = ah_2 + ak_0 + k_1$, establishing the displayed deconvolution. Common-rate kernels satisfy $k_0 = \eta Q > 0$ and $k_1 = (1 - \eta)k_0$. Solving these two identities gives (S13). □

Consequently, "unknown rate" is not a permanent information restriction when these additional exact aggregate lags are observed. Finite-sample recovery can nevertheless be extremely ill-conditioned when k_0 is small. At $Q = 0$, group weights are zero but the memory rate is not identified.

S4.5 Known heterogeneous modes

For distinct known rates η_i, the first n kernel moments satisfy

$$(k_0, \ldots, k_{n-1})^\top = V_\eta q, \qquad (V_\eta)_{j+1,i} = \eta_i(1 - \eta_i)^j.$$

This is a Vandermonde matrix times a positive diagonal matrix. Its determinant is a nonzero product of η_i and pairwise differences of $1 - \eta_i$, so exact recovery holds. Close rates can make this inversion poorly conditioned. Equal-rate cohorts reveal only their total weight through this observation law.

S5 Finite-sample response ellipsoids with the correct calibration

S5.1 Sampling model

Let $p = H + 1$ be a predeclared response dimension. For R independent episodes,

$$Y_r = m + u_r\beta + \varepsilon_r, \qquad \varepsilon_r \overset{\text{iid}}{\sim} N_p(0, \Sigma), \quad \Sigma \succ 0, \tag{S14}$$

conditional on a nonconstant probe design independent of the residuals. Here $m \in \mathbb{R}^p$ is an arbitrary common no-probe path. Define

$$v_r = u_r - \bar{u}, \quad D = \sum_r v_r^2 > 0, \quad \nu = R - 2,$$

$$\hat\beta = \frac{\sum_r v_r Y_r}{D}, \quad \hat m = \bar Y - \bar u\hat\beta, \quad \hat\Sigma = \frac{1}{\nu}\sum_r (Y_r - \hat m - u_r\hat\beta)(Y_r - \hat m - u_r\hat\beta)^\top, \qquad V = \hat\Sigma/D.$$

The actual sampling covariance of $\hat\beta$ is Σ/D, not the random estimate V.

Theorem S5.1 (Exact response ellipsoid)**.** *If $\nu \ge p$, set*

$$k_\alpha^2 = \frac{\nu p}{\nu - p + 1} F^{-1}_{p,\nu-p+1}(1-\alpha), \qquad \mathcal{E}_\alpha = \left\{ b : (b-\hat\beta)^\top V^{-1}(b-\hat\beta) \le k_\alpha^2 \right\}. \tag{S15}$$

Then $\mathbb{P}(\beta \in \mathcal{E}_\alpha \mid u) = 1 - \alpha$.

Proof. The design matrix has columns $\mathbf{1}$ and u and rank two. Gaussian least-squares projection yields

$$\hat\beta - \beta \sim N_p(0, \Sigma/D), \qquad \nu\hat\Sigma \sim W_p(\Sigma, \nu),$$

independently, conditional on the design. To see the independence directly, transform the R episode coordinates by an orthogonal matrix whose first two rows span the design. The remaining ν rows are independent Gaussian residual coordinates. The fitted slope is a linear combination of the first two coordinates, independent of all remaining rows.

Whiten the slope error and residual covariance:

$$z = \sqrt{D}\,\Sigma^{-1/2}(\hat\beta - \beta) \sim N_p(0, I), \qquad W = \nu\Sigma^{-1/2}\hat\Sigma\Sigma^{-1/2} \sim W_p(I, \nu).$$

They are independent. Therefore

$$T^2 = (\hat\beta - \beta)^\top V^{-1}(\hat\beta - \beta) = \nu z^\top W^{-1} z.$$

By rotational invariance, rotate the direction of z to the first coordinate. Its squared norm is $U \sim \chi^2_p$. Write $W = G^\top G$ for a $\nu \times p$ standard Gaussian matrix and partition its first column g_1 from the remaining columns G_2. The inverse-entry identity gives

$$(W^{-1})_{11} = \left[g_1^\top (I - P_{G_2}) g_1 \right]^{-1}.$$

The projector has rank $p-1$ almost surely, so the denominator has distribution $Z \sim \chi^2_{\nu-p+1}$. The rotationally transformed W has its original law and is independent of $\|z\|$; hence U, Z are independent. Thus

$$T^2 \overset{d}{=} \nu U/Z, \qquad \frac{\nu - p + 1}{\nu p} T^2 \sim F_{p,\nu-p+1}.$$

The quantile in (S15) proves the claim. This is a specialization of classical Hotelling inference [3, 4]. □

Important limitations. If $\nu < p$, the full residual covariance is singular almost surely under this sampling law. A pseudoinverse or shrinkage estimator does not inherit the theorem. Use the retained coordinatewise method, a justified fixed subspace model, or another separately proved construction. For heteroskedastic, dependent, selected, or non-Gaussian episodes, the exact F calibration is not established here. A sandwich covariance generally needs its own asymptotic or finite-sample argument.

S5.2 What the ellipsoid support formula actually proves

Proposition S5.2 (Linear contrast support)**.** *For $V \succ 0$, $k \ge 0$, and fixed c, the extrema of $c^\top b$ over $(b-\hat\beta)^\top V^{-1}(b-\hat\beta) \le k^2$ are*

$$c^\top\hat\beta \pm k\sqrt{c^\top V c}.$$

Proof. Write $b = \hat{\beta} + V^{1/2}z$, $\|z\|_2 \le k$. Cauchy–Schwarz gives $c^\top b \le c^\top \hat{\beta} + k \left\| V^{1/2}c \right\|_2$. For $c \neq 0$, equality holds at $z = kV^{1/2}c / \left\| V^{1/2}c \right\|_2$. Negating c gives the minimum. If $c = 0$, both extrema are zero. □

This is standard ellipsoid geometry [5]. It is not a formula for an unidentified structural parameter.

S5.3 Why no fixed response contrast gives S

Proposition S5.3 (No universal finite-response formula)**.** *On the manuscript's unrestricted finite positive-memory family, S is not generally a function of a fixed finite public-response vector. In particular, it cannot generally equal $c^\top \beta$ for a fixed vector c.*

Proof. The retained finite-window theorem supplies two models with the same response vector over any prescribed finite horizon and opposite stability classes [1, Theorem 8.2]. A function of that common vector has the same value in both models, whereas their S values differ. This is a contradiction.

For a concrete four-coefficient witness, use $a = 2/5$, rates $(1, 1/2, 1/20)$, and weights

$$q^{(0)} = (0, 12/65, 0), \qquad q^{(1)} = (54/1235, 0, 240/247).$$

Both normalized responses begin

$$(h_0, h_1, h_2, h_3) = (1, 2/5, 82/325, 23/125),$$

but $S_0 = 38/65 < 1$ and $S_1 = 92/65 > 1$. The impact normalization may be chosen the same in both models. □

Under stronger fixed-order assumptions a nonlinear recovery can exist. For the common single-mode aggregate with $k_0 > 0$,

$$S = a + \frac{k_0^2}{k_0 - k_1},$$

using Proposition S4.3. This is still not a linear contrast of the response vector, and its instability near a small denominator requires uncertainty propagation.

S5.4 The audited contrast is a different problem

Under I1, let $\theta = (\mu, a, q_1, \ldots, q_d)^\top$ be the coefficient vector in the manuscript's terminal-row regression. Then $S = c^\top \theta$ with $c = (0, 1, \ldots, 1)^\top$. Conditional on a full-rank design X, independent Gaussian terminal errors of common unknown variance give

$$\frac{c^\top \hat{\theta} - c^\top \theta}{\hat{\sigma}\sqrt{c^\top (X^\top X)^{-1} c}} \sim t_{N - d_\theta}.$$

Indeed, the numerator is Gaussian, the independent residual sum of squares divided by the true variance is chi-squared with $N - d_\theta$ degrees of freedom, and their ratio is Student-t. This is the retained classical I1 pivot, not a new solution of I0. A full simultaneous coefficient ellipsoid may answer simultaneous-contrast questions, but its support does not automatically improve a prespecified one-contrast interval.

S6 The exact covariance-aware identified set

Let $\underline{\eta} > 0$ be an externally valid rate floor. Let μ be a finite nonnegative measure on $[\underline{\eta}, 1]$; in a finite-state model it is $\sum_i q_i \delta_{\eta_i}$. Define $\psi_j(\eta) = \eta(1-\eta)^j$. For a predeclared persistence bound $0 \le a \le \bar{a} \le 1$, define

$$\mathcal{K}_\alpha = \Big\{ (a, \mu, b) : b \in \mathcal{E}_\alpha, \quad b \ge 0, \quad b_0 > 0, \quad 0 \le a \le \bar{a}, \quad \mu \ge 0, \\ b_1 = a b_0, \quad b_{t+1} = a b_t + \sum_{j=0}^{t-1} b_{t-1-j} \int \psi_j(\eta) d\mu(\eta), \quad 1 \le t \le H-1 \Big\}. \tag{S16}$$

The structural interval hull is

$$S^-_{\text{exact}} = \inf_{\mathcal{K}_\alpha} (a + \mu([\underline{\eta}, 1])), \qquad S^+_{\text{exact}} = \sup_{\mathcal{K}_\alpha} (a + \mu([\underline{\eta}, 1])). \tag{S17}$$

The set of attainable S need not itself be an interval; (S17) is its smallest enclosing interval. Choosing $\bar{a} = 1$ uses a conservative closed extension of the manuscript's $a < 1$ family.

Theorem S6.1 (Coverage through model compatibility)**.** *Suppose the true model obeys the declared parameter class and sampling contract. On $\{\beta \in \mathcal{E}_\alpha\}$, its true (a, μ, β) belongs to (S16). Therefore any interval enclosing all objectives in (S16) covers the true S with probability at least $1 - \alpha$.*

Proof. The true response is nonnegative and satisfies the recurrence by construction. Its rates and persistence satisfy the declared class. The ellipsoid event supplies the remaining membership requirement. Thus the true objective belongs to the enclosed range on an event of probability $1 - \alpha$. □

This simple containment argument is what allows a rigorous relaxation. It does not establish that the exact nonlinear set is convex: the products $a b_t$ and $b_s \int \psi_j d\mu$ are bilinear.

S7 A finite, covariance-aware conic outer relaxation

This section supplies a concrete alternative to the incorrect closed-form claim. It uses a second-order cone constraint for the response ellipsoid and linear outer constraints for the recurrence. McCormick inequalities are established convex-relaxation machinery [6]; the continuum-support containment argument is proved here.

S7.1 Response bounds and a finite feedback bound

The ellipsoid's coordinate projections give

$$L_s = \max\{0, \hat{\beta}_s - k_\alpha \sqrt{V_{ss}}\}, \qquad U_s = \hat{\beta}_s + k_\alpha \sqrt{V_{ss}}.$$

If some $U_s < 0$, no nonnegative response is compatible. In the construction below require $H \ge 2$ and $L_0 > 0$.

Lemma S7.1 (Data-derived finite mass cap)**.** *Every model in (S16) satisfies*

$$Q \le M := \frac{U_2}{\underline{\eta} L_0}. \tag{S18}$$

Proof. The lag-2 recurrence is $b_2 = ab_1 + b_0 \int \eta \, d\mu$. Positivity gives $b_2 \geq b_0 \underline{\eta} Q \geq L_0 \underline{\eta} Q$, while $b_2 \leq U_2$. Rearrangement proves the bound. □

If $L_0 \leq 0$, this particular construction has not obtained a finite cap. Return an inconclusive status or use a separately justified external $Q_{\max}$; do not introduce an arbitrary fitting cap and call it learned information. Empty ellipsoid/model intersections are conflicts, not successful stability certificates.

S7.2 Continuous rate support without a grid assumption

Partition $[\underline{\eta}, 1]$ into bins B_ℓ, $\ell = 1, \ldots, L$. Use a nonoverlapping membership convention and closed-bin envelopes at boundaries. Set

$$w_\ell = \mu(B_\ell), \quad \ell_{j\ell} = \min_{\eta \in \overline{B}_\ell} \psi_j(\eta), \quad u_{j\ell} = \max_{\eta \in \overline{B}_\ell} \psi_j(\eta).$$

The minimum is at a bin endpoint; the maximum is at an endpoint or the projection of $1/(j+1)$ into the bin. This follows by differentiating $\eta(1-\eta)^j$. Consequently the envelopes cover off-grid rates. They do not assert that actual memory rates lie at chosen grid nodes.

S7.3 Bilinear-product lifting

Introduce variables v_t for ab_t, and $z_{\ell s}$ for $w_\ell b_s$. The exact equalities are relaxed by four inequalities. For $0 \leq x \leq A$, $L \leq y \leq U$, a valid product $z = xy$ satisfies

$$z \geq Lx, \qquad z \geq Ay + Ux - AU, \tag{S19}$$

$$z \leq Ay + Lx - AL, \qquad z \leq Ux. \tag{S20}$$

Each inequality follows by expanding one of the nonnegative products $x(y-L)$, $(A-x)(U-y)$, $(A-x)(y-L)$, or $x(U-y)$. Use (S19)–(S20) with $(x, y, z, A) = (a, b_t, v_t, \bar{a})$ and $(w_\ell, b_s, z_{\ell s}, M)$, respectively.

S7.4 The conic program

Let $V = RR^\top$ be a positive-definite factorization. The outer set $\mathcal{R}_\alpha$ has variables a, w, b, v, z and constraints:

$$\left\| R^{-1}(b - \hat{\beta}) \right\|_2 \leq k_\alpha, \qquad L \leq b \leq U, \tag{S21}$$

$$0 \leq a \leq \bar{a}, \quad w_\ell \geq 0, \quad \sum_\ell w_\ell \leq M, \tag{S22}$$

all product envelopes (S19)–(S20),

$$b_1 = v_0, \tag{S23}$$

$$v_t + \sum_{\ell=1}^{L} \sum_{j=0}^{t-1} \ell_{j\ell} z_{\ell,t-1-j} \leq b_{t+1} \leq v_t + \sum_{\ell=1}^{L} \sum_{j=0}^{t-1} u_{j\ell} z_{\ell,t-1-j}, \quad 1 \leq t \leq H-1. \tag{S24}$$

The required product variables use only the response coordinates appearing in those formulas. Their finite bounds follow from $0 \leq v_t \leq \bar{a} U_t$ and $0 \leq z_{\ell s} \leq M U_s$. Define

$$S^-_{\text{soc}} = \min_{\mathcal{R}_\alpha} \left(a + \sum_\ell w_\ell \right), \qquad S^+_{\text{soc}} = \max_{\mathcal{R}_\alpha} \left(a + \sum_\ell w_\ell \right). \tag{S25}$$

These are convex conic problems: a linear objective, affine inequalities/equalities, and a second-order cone constraint. Maximization is implemented by minimizing the negative linear objective. Global convexity of the relaxation is not exactness of the original bilinear problem.

Theorem S7.2 (True-model containment and statistical validity)**.** *Every model in* (S16) *lifts into* $\mathcal{R}_\alpha$ *with the same value of* S*. For nonempty sets,*

$$S^-_{\mathrm{soc}} \le S^-_{\mathrm{exact}} \le S^+_{\mathrm{exact}} \le S^+_{\mathrm{soc}}. \tag{S26}$$

Under Theorem S5.1's sampling law, exact or outward-valid endpoints in (S25) *cover the true* S *with probability at least* $1-\alpha$*.*

Proof. For a true feasible (a, μ, b), set $w_\ell = \mu(B_\ell)$, $v_t = ab_t$, and $z_{\ell s} = w_\ell b_s$. The mass bound follows from Lemma S7.1. Exact products satisfy all four envelopes. The first lag holds exactly. For every j, ℓ, s, since $b_s \ge 0$ and $\mu \ge 0$,

$$\ell_{j\ell} w_\ell b_s \le b_s \int_{B_\ell} \psi_j(\eta) d\mu(\eta) \le u_{j\ell} w_\ell b_s.$$

Summing these inequalities in the exact recurrence gives (S24). The response and its bounds satisfy (S21). Finally $a + \sum_\ell w_\ell = a + \mu([\underline{\eta}, 1])$. Containment gives the ordering of extrema. Theorem S6.1 then gives the probability statement. The mass cap is data-dependent but valid pointwise on the same ellipsoid event, so deriving it this way does not consume a separate error budget. □

S7.5 A correct comparison with a rectangle

Let the comparison rectangle be the coordinate enclosure $[L, U]$ generated from this *same* ellipsoid. Define the paper-style box LP using

$$aL_t + \sum_{\ell, j<t} \ell_{j\ell} L_{t-1-j} w_\ell \le U_{t+1}, \tag{S27}$$

$$aU_t + \sum_{\ell, j<t} u_{j\ell} U_{t-1-j} w_\ell \ge L_{t+1}, \tag{S28}$$

including the empty-sum first lag and the same persistence and nonnegative-weight bounds.

Theorem S7.3 (Weak dominance for a common underlying region)**.** *Every conic-feasible point projects onto a feasible point of* (S27)–(S28)*. Therefore its structural interval is no wider:*

$$S^-_{\mathrm{box}} \le S^-_{\mathrm{soc}} \le S^+_{\mathrm{soc}} \le S^+_{\mathrm{box}}.$$

No strict improvement and no dominance over a separately calibrated Bonferroni rectangle are asserted.

Proof. The product envelopes imply $aL_t \le v_t \le aU_t$ and $w_\ell L_s \le z_{\ell s} \le w_\ell U_s$. Consequently the left side of (S27) is at most $v_t + \sum \ell_{j\ell} z_{\ell, t-1-j}$, which is at most $b_{t+1} \le U_{t+1}$. The analogous upper expression is at least $b_{t+1} \ge L_{t+1}$. Thus the same (a, w) is box-feasible, with the same objective. Taking extrema proves the result. □

A known-covariance two-dimensional counterexample explains the qualification. For identity covariance and nominal 95% coverage, the chi-squared ellipsoid radius is $\sqrt{\chi^2_{2,0.95}} \approx 2.44775$, while the Bonferroni coordinate half-width is $\Phi^{-1}(1 - 0.05/4) \approx 2.24140$. The ellipsoid is wider for the first-coordinate contrast, but narrower for the $(1, 1)$ contrast, whose half-widths are

approximately 3.46164 and 4.48281. Equal confidence level or smaller volume is not set inclusion. Intersecting two independently calibrated regions requires a joint coverage argument, such as splitting the error budget; one cannot keep both original 95% claims and silently treat their intersection as 95%.

S8 Convergent refinement toward the exact compatible interval

A single conic relaxation can remain loose. This section states a rigorous but potentially expensive sharpening procedure. It is not a claim that refining rate bins alone removes product uncertainty.

S8.1 Compact target and finite support

Fix the observed ellipsoid, global bounds L, U, $L_0 > 0$, finite M, and $a \in [0, \bar{a}]$. Let $\mathcal{K}$ be (S16) with $Q \le M$. It is a compact feasible set when measures are given the weak topology on the compact rate interval: bounded nonnegative measures are sequentially compact, and all finite recurrence integrands are continuous. The constraint $b_0 \ge L_0 > 0$ removes the open-boundary issue.

The measure formulation need not enlarge the finite-response unknown-order model. For fixed $b, a, Q > 0$, only the first $H-1$ moments of ψ_j, $j = 0, \dots, H-2$, enter the recurrence. The normalized moment vector lies in the convex hull of the compact curve $\eta \mapsto (\psi_0(\eta), \dots, \psi_{H-2}(\eta))$ in $\mathbb{R}^{H-1}$. By the finite-dimensional convex-hull representation, it is a convex combination of at most H points. Multiplying by Q supplies an atomic measure with at most H rates, preserving the response and objective. At $Q = 0$ the zero measure suffices. This does not apply to a separately fixed smaller state-count class.

S8.2 Exhaustive refinement theorem

Partition the global response box into finitely many subboxes of maximum coordinate width $\epsilon_n \to 0$, and partition the rate interval into bins of maximum diameter $\delta_n \to 0$. For each response subbox, intersect it with the same ellipsoid and solve the outer program using that subbox's response bounds and the fixed global mass cap M. Let L_n be the smallest lower endpoint over all feasible subboxes and U_n the largest upper endpoint. Infeasible nodes may be discarded only with a valid infeasibility certificate.

Theorem S8.1 (Convergence of exhaustive outer intervals)**.** *Assume $\mathcal{K} \neq \varnothing$. Under exact solution of every node's convex outer program,*

$$L_n \longrightarrow \min_{\mathcal{K}} S, \qquad U_n \longrightarrow \max_{\mathcal{K}} S.$$

Taking the running maximum of lower bounds and running minimum of upper bounds makes the reported enclosure monotone. This theorem gives no favorable computational complexity bound.

Proof. Every exact feasible response lies in at least one subbox, so Theorem S7.2 gives $L_n \le \min_{\mathcal{K}} S \le \max_{\mathcal{K}} S \le U_n$.

Choose outer-feasible maximizers along any subsequence that attains the limit superior of U_n. Denote their variables by a_n, b_n, w_n, v_n, z_n. Choose one representative rate $\eta_{\ell,n}$ in each bin and form $\mu_n = \sum_\ell w_{\ell,n} \delta_{\eta_{\ell,n}}$. These measures have mass at most M. Compactness yields a subsequence with $a_n \to a$, $b_n \to b$, and $\mu_n \Rightarrow \mu$ weakly. The limit response is in the fixed closed ellipsoid and its global box.

Inside a response subbox of width at most ϵ_n, both $z_{\ell s,n}$ and $w_{\ell,n}b_{s,n}$ lie in $[w_{\ell,n}L_{s,n}, w_{\ell,n}U_{s,n}]$. Hence

$$|z_{\ell s,n} - w_{\ell,n}b_{s,n}| \le w_{\ell,n}\epsilon_n, \qquad |v_{t,n} - a_n b_{t,n}| \le \bar{a}\epsilon_n.$$

Let $B_* = \max_s U_s$ and let $\omega(\delta)$ be the maximum oscillation of the finitely many functions ψ_j, $0 \le j \le H-2$, on rate intervals of diameter at most δ. Uniform continuity gives $\omega(\delta) \to 0$. Since $0 \le \psi_j \le 1$, the difference between the relaxed right side and the exact recurrence evaluated at (a_n, b_n, μ_n) is, for lag t, at most

$$(\bar{a} + tM)\epsilon_n + tMB_*\omega(\delta_n).$$

This tends to zero uniformly over the finitely many lags. Weak convergence preserves each continuous kernel moment, so the limiting (a, b, μ) satisfies every exact recurrence and all class bounds. Thus it belongs to $\mathcal{K}$.

The objective is continuous along this subsequence because total mass is the integral of the continuous function one: $a_n + \mu_n([\underline{\eta}, 1]) \to a + \mu([\underline{\eta}, 1]) \le \max_{\mathcal{K}} S$. Therefore $\limsup U_n \le \max_{\mathcal{K}} S$. Containment gives the reverse inequality. Apply the same argument to minimizing sequences to obtain convergence of L_n. Running intersections retain validity and the same limits. □

Practical consequence. A branch-and-bound implementation may prioritize useful subboxes, but it must retain bounds for every unexplored node. Stopping after promising nodes and ignoring others destroys the global guarantee. Rate refinement and response-product refinement solve different sources of relaxation error.

S9 Numerical endpoint certification, not just optimizer success

Convexity removes local-versus-global ambiguity at the mathematical optimization level. It does not make floating-point endpoint reports outward-valid. A feasible primal point for a minimization is an upper bound on its optimum, not the lower endpoint needed for certification. The analogous direction reverses for maximization.

Proposition S9.1 (Residual-corrected conic lower bound)**.** *Consider*

$$p_* = \min\{c^\top x : Ax = b,\ h - Gx \in K,\ \ell \le x \le u\},$$

where K is a closed convex cone and all coordinate bounds are finite. For arbitrary y and any exactly dual-cone-feasible $z \in K^*$, put $r = c + A^\top y + G^\top z$. Then

$$p_* \ge -y^\top b - z^\top h + \sum_i \min\{r_i\ell_i, r_i u_i\}. \tag{S29}$$

Negating the objective and the resulting lower bound supplies an upper bound for a maximization problem.

Proof. For a feasible x, $y^\top(Ax - b) = 0$ and $z^\top(Gx - h) = -z^\top(h - Gx) \le 0$. Therefore

$$c^\top x \ge c^\top x + y^\top(Ax - b) + z^\top(Gx - h) = r^\top x - y^\top b - z^\top h.$$

The minimum of the linear form $r^\top x$ on a box is the sum of its coordinate minima, giving (S29). This argument requires no exact stationarity $r = 0$. □

Use exact rational comparisons or outward interval arithmetic for dual-cone membership and the lower-bound evaluation. The second-order cone is self-dual; membership of (t, v) can be verified

from $t \geq 0$ and $t^2 \geq v^\top v$ with directed arithmetic. Covariance factorization and kernel-envelope coefficients also need numerical enclosures when claiming a formally enclosed endpoint. A library's success flag, a small residual, or a primal objective alone is insufficient. Conic duality is established optimization theory [5]; this proposition spells out a bounded-domain numerical safeguard.

A production result should separately store: primal feasibility, dual-cone feasibility, numerical enclosure error, optimization gap, statistical confidence level, and structural-contract validity. Failure at any required stage means inconclusive or conflict, never automatic stability.

Implementation status. The released numerical routine now retains the complete solved tangent matrices and applies residual correction to both objectives, using the negative-objective lower bound for the upper endpoint. It records solver and cut-termination diagnostics separately. These improvements do not enclose floating-point construction of the covariance geometry, mass cap, kernel envelopes, or cuts. Its endpoint candidates are therefore diagnostic; the structural certification flag remains false and certified classification is withheld. The theorem above is conditional on outward-valid arithmetic, not a proof that this numerical routine has attained it.

S10 What coverage guarantees, and why abstention cannot disappear

Theorem S10.1 (Directionally valid classifications)**.** *Suppose an interval $[L_S, U_S]$ covers the true S with probability at least $1-\alpha$. Issue STABLE if $U_S < 1$, UNSTABLE if $L_S > 1$, and ABSTAIN otherwise; an empty or unverified set is a separate non-certificate status. Then the unconditional probability of a wrong strict label is at most α. If the probability of issuing a strict label is $c > 0$, the implied conditional bound is at most $\min\{1, \alpha/c\}$, not automatically α.*

Proof. On the coverage event, $U_S < 1$ implies $S < 1$, and $L_S > 1$ implies $S > 1$. Thus any wrong strict label is contained in the noncoverage event. Divide its probability by c for the conditional statement. When $c = 0$, conditional error is undefined. At $S = 1$ either strict label is wrong. □

Theorem S10.2 (Two-world abstention bound)**.** *Let P_0, P_1 be observation laws for opposite strict stability classes. Under equal class weights, let c be the probability of issuing either strict label and e the unconditional wrong-label probability. Then*

$$c \leq \min\{1,\ 2e + \mathrm{TV}(P_0, P_1)\}.$$

If the two laws are identical, $c = 2e$. For equal-covariance Gaussian laws with Mahalanobis mean distance d, the bound is

$$c \leq \min\{1,\ 2e + 2\Phi(d/2) - 1\}.$$

Proof. Let A_0, A_1 be the disjoint decision events labeling the respective classes. Then

$$\begin{aligned} c &= \tfrac{1}{2}\{P_0(A_0) + P_0(A_1) + P_1(A_0) + P_1(A_1)\}, \\ e &= \tfrac{1}{2}\{P_0(A_1) + P_1(A_0)\}. \end{aligned}$$

Subtracting gives

$$c - 2e = \tfrac{1}{2}\{P_0(A_0) - P_1(A_0) + P_1(A_1) - P_0(A_1)\} \leq \mathrm{TV}(P_0, P_1).$$

When $P_0 = P_1$, the difference is zero. For equal-covariance Gaussians, whitening and projection onto the mean-difference direction reduce the optimal likelihood-ratio test to two univariate unit-variance normal distributions separated by d. Their midpoint threshold has error $\Phi(-d/2)$, so their total variation is $1 - 2\Phi(-d/2) = 2\Phi(d/2) - 1$. □

No ellipsoid or optimizer can remove this information limitation. It can reduce an avoidable relaxation loss while leaving unavoidable uncertainty intact. In particular, the manuscript's displayed level-noise forced error of 47.56% and error target 5% give a decision-coverage bound near 14.88%, not 9.75%.

S11 Financial safety and empirical usefulness are separate statements

S11.1 A precise solvency non-transfer example

Proposition S11.1 (No solvency guarantee from a stable signal alone)**.** *Fix any scalar signal persistence $r \in [0, 1)$ and any $\pi \in [0, 1]$. There is a finite-price, initially solvent account and an external-shock law with signal transition coefficient r for which insolvency at a specified horizon has probability π.*

Proof. Take price $P_t = P_0 + x_t$ with $x_{t+1} = rx_t + \varepsilon_{t+1}$, initially $x_0 = 0$. Let a fixed account hold $I > 0$ shares and have equity $E_0 > 0$; choose $P_0 > 2E_0/I$ and cash $C = E_0 - IP_0$. The initial equity is $C + IP_0 = E_0$. Assume funding permits this initial position; the proposition specifies no particular leverage or credit rule. Set all innovations before the designated time to zero. At that time let the innovation be $-2E_0/I$ with probability π and zero otherwise. On the shock event the price remains positive but the equity becomes $E_0 + I(-2E_0/I) = -E_0$. Otherwise it remains E_0. The signal coefficient is still $r < 1$, while insolvency probability is exactly π. □

This is a scope counterexample, not a new liquidity-spiral theory. Its purpose is to show that shock distributions, exposures, collateral, and execution constraints cannot be inferred from a stable eigenvalue. The manuscript's funded simulator must still measure its own predeclared scenario probabilities and mechanism contrasts.

S11.2 Why a theoretical solution does not finish the empirical study

The response ellipsoid proves coverage only under (S14). It does not prove that one historical market stream supplies iid reset episodes. The continuous-time theorem does not prove that actual agents obey the model. The group theorem does not show which real accounts are adaptive. A new interval construction does not prove an improvement over the original method until both are evaluated with matched inputs, coverage targets, and numerical safeguards.

Similarly, computing features on Period A and fitting a regression using Period B outcomes does not by itself measure forecasting performance of a rule frozen before those B outcomes were seen. A later untouched date set remains necessary for that particular predictive claim. Missing historical spread or depth observations must be removed from the corresponding primary analysis or explicitly restricted to a retrospective diagnostic.

The correct remaining empirical work is to report interval coverage, widths, decision exposure, conditional and unconditional errors, model conflicts, runtime, and actual forward predictive loss under a frozen protocol. A null result, noncomparability finding, or frequent abstention is admissible. No proof here guarantees a grade, acceptance, a target p-value, or a fixed percentage of project completion.

S12 Integration into the manuscript

Manuscript location	Insert or replace	What not to claim
Chapters 2 & 3	Theorem S3.3, Proposition S3.5, and Euler theorem.	Do not say $\gamma = 1 - a$ is a limit without scaling by a physical step. Do not claim arbitrary Euler steps preserve stability.
Chapter 5 & Appendix A	Theorems S4.1, S4.2, Proposition S4.3.	Do not say an unknown rate always leaves a ray when known mass or extra aggregate lags remove that ambiguity.
Chapter 4	Theorem S5.1 and covariance-rank failure status.	Do not replace the exact F radius with a chi-squared radius when covariance is estimated.
Chapter 4	Exact compatible set, Theorem S7.2, and common-region comparison.	Do not call S a linear response contrast or promise strict improvement over the original Bonferroni method.
Technical appendix	Theorem S8.1 and Proposition S9.1.	Do not treat numerical local fits or unchecked duals as globally enclosed endpoints.
Chapter 4	Matched 120-cell benchmark (Table 5) and budget sweep (Figure 4).	Do not extrapolate decisive classification from large budgets to under-sampled settings.
Chapter 5	Retain the fixed-contrast I1 Student-t pivot (Table 9).	Do not count standard ellipsoid support as a replacement advance over the existing single-contrast result.
Chapter 6	Preserve signal/solvency decoupling and paired mechanism controls (Tables 11–13).	Do not attribute inventory buffer halts to balance-sheet insolvency.
Chapter 7	Exploratory digital asset associations and Gates 2–4 calibration (Tables 14–17).	Do not substitute observational associations for locked predictive tests or proprietary runtime audits.
Abstract / conclusion	Describe standard tools, specialized derivations, and executed evidence separately.	Do not state that all missing gaps, all conditional errors, or all abstentions are eliminated.

S12.1 Replacement contribution paragraph

We separate the observations needed for aggregate feedback inference from those needed for componentwise attribution and financial-risk evaluation. In the positive-feedback class, a heterogeneous-rate continuous-time analogue admits a spectral small-gain characterization, with the original discrete boundary recovered under an explicit Euler scaling. For independent Gaussian probe episodes, an exactly calibrated response ellipsoid accounts for estimated within-episode covariance. Rather than treating aggregate feedback as a linear contrast of finite responses, we impose the positive-memory recurrence and construct a conic outer bound that preserves compatible continuous-rate models. The bound is no wider than the recurrence box relaxation derived from the same ellipsoid, while improvement over separately

calibrated rectangular methods remains an empirical question. Group attribution depends on the rank and normalization information of the actual experimental map, and none of these signal-level results alone certifies funded-system safety.

This statement coordinates the mathematical scope with the empirical evaluations executed in the main manuscript (Chapter 4, Table 5, Figure 4).

S12.2 Implementation procedure

1. Freeze the response dimension, probe design, observation contract, rate floor, and error level before confirmation data are examined.
2. Compute the OLS response, residual covariance, residual degrees of freedom, and exact F critical value. Reject invalid rank or insufficient sample-size cases.
3. Form the ellipsoid, its nonnegative coordinate enclosure, and the finite mass cap. Record weak-impact or model-conflict status separately.
4. Build continuous rate-bin envelopes and the lifted conic recurrence program. Do not replace continuous support with a bare grid without a declared approximation.
5. Solve both endpoint problems; retain primal, dual, residual, and gap information. Return certificates only from outward-valid endpoint bounds.
6. Optionally refine response subboxes and rate bins while retaining valid bounds for every unresolved region.
7. Score the new and retained methods on identical I0 panels. Keep I1 comparisons separate or explicitly label their stronger information.
8. Generate every table and claim from saved result rows. Preserve unsuccessful solves and undefined conditional-error denominators.

S12.3 Required comparison table

For each method and generating setting, record the number of independent panels, nominal confidence, observed coverage with Monte Carlo uncertainty, median/quantile interval widths, stable/unstable/abstain/conflict counts, unconditional error, conditional error, solver failures, runtime, and probe cost. Also retain covariance dimension, residual degrees of freedom, rate-bin resolution, numerical enclosure status, and the exact information contract.

The minimum comparison is: the original Bonferroni recurrence LP, the enclosing-box LP derived from the new ellipsoid, the conic relaxation, and, if implemented, its refinement. This separates a change of confidence region from a change of recurrence relaxation. The separately audited I1 regression is an information benchmark, not an equal-information competitor.

S13 Executed checks and what they do not establish

The reference directory contains 21 unit tests. Looped numerical fixtures also cover 120 continuous systems with heterogeneous rates, common-rate complex modes, Euler behavior, group recovery, finite-window witnesses, covariance support, rate envelopes, and decision-bound arithmetic. These are finite numerical consistency checks, not proofs of the universal statements.

A Gaussian calibration illustration used $R = 32$, $p = 4$, 10,000 independently generated panels, and seed 240926. The exact response-ellipsoid radius squared was 12.12340136. The true response was included in 9,476 panels (94.76%), with nominal Monte Carlo standard error about 0.218 percentage points at 95%. Replacing its F radius by an unjustified χ^2_4 radius gave 89.46%

inclusion in this particular illustration. This is a diagnostic example, not evidence about the market-data study.

The polyhedral outer relaxation of the Second-Order Cone Program (Theorem S7.2) is implemented via HiGHS cutting planes with numerical dual residual repair (`research_extension.mm_revision.inference.conic_certificate.fit_conic_certificate`). The numerical comparisons presented represent the validated configurations. The corrected 600-panel grid starts at the post-probe response. Two separate 300-panel budget sweeps distinguish known covariance with chi-square calibration from independent Wishart-estimated covariance with F calibration. The main manuscript reports their candidate endpoints and failures from `research_extension/outputs/statistical_repair_20260926/`. Full outward construction of the numerical constraints remains unverified, so all conic certified labels are withheld. The interval-containment theorem and the solver's numerical evidence are distinct claims.

The calculation log also records the manuscript's $S = 0.82$, spectral radius 0.9900902493, and the conventional two-sided value $\mathbb{P}(|T_8| \geq 2.47) \approx 0.03871045$. Actual regression tables should still be regenerated from their saved model objects rather than edited using this isolated calculation.

```
# Run from the reference directory after installing NumPy and SciPy.
python -m unittest discover -v
python run_checks.py
```

S14 Executed Multi-Agent Continuous Reinforcement Learning Suite

Status: fully executed across 10 random seeds in PyTorch 2.5.1 on continuous state space. This section details the complete mathematical formulation, network architectures, and empirical findings for evaluating feedback stability in multi-agent reinforcement learning (MARL) populations and continuous trading environments.

S14.1 Continuous State Space and Balance-Sheet Conservation

The continuous trading environment (`continuous_trading_env.py` in `research_extension/learned_agents/`) formalizes the interaction of learning agents with a double-auction price-formation mechanism under strict balance-sheet constraints. At discrete decision step t, the state of the market and the agent is represented by the continuous vector $\mathbf{s}_t \in \mathbb{R}^5$:

$$\mathbf{s}_t = (x_t,\, r_t,\, I_t,\, \mathrm{util}_t,\, \mathrm{imb}_t)^\top \in \mathbb{R}^5, \tag{S30}$$

where:

- $x_t \in \mathbb{R}$ is the normalized price displacement from fundamental anchor value $p^* = 100.0$, i.e., $x_t = P_t - p^*$;
- $r_t = (P_t - P_{t-1})/P_{t-1} \in \mathbb{R}$ is the instantaneous log-price return;
- $I_t \in [-I_{\max}, I_{\max}]$ is the agent's net inventory position (with $I_{\max} = 10.0$ units);
- $\mathrm{util}_t = I_t / I_{\max} \in [-1, 1]$ is the normalized inventory capacity utilization;
- $\mathrm{imb}_t = (V_t^b - V_t^a)/(V_t^b + V_t^a) \in [-1, 1]$ is the top-of-book depth imbalance.

Every executed order transfers cash and shares between counterparties according to exact double-entry conservation laws:

$$C_{t+1} = C_t - P_t a_t - \kappa_{\mathrm{fee}} |a_t| P_t, \qquad I_{t+1} = I_t + a_t, \tag{S31}$$

where $a_t \in [-a_{\max}, a_{\max}]$ is the executed trade action, C_t is the cash balance, and $\kappa_{\text{fee}} = 5\times10^{-4}$ (5 bps) is the exchange trading fee. Across all transitions and evaluation episodes, cash-conservation and inventory-conservation residuals are verified numerically to satisfy:

$$|\Delta C_t + P_t a_t + \kappa_{\text{fee}}|a_t|P_t| \le 1.0 \times 10^{-11}, \qquad |\Delta I_t - a_t| \le 1.0 \times 10^{-11}. \tag{S32}$$

S14.2 Neural Deep RL Architectures: Discrete DQN and Continuous Actor-Critic

We evaluate two distinct neural deep reinforcement learning architectures trained across 10 independent random seeds (926101–926110) for 50,000 transitions each (`research_extension/learned_agents/train_deep_rl.py`):

1. **Deep Q-Network (DQN):** Fully connected multilayer perceptron $f_\theta : \mathbb{R}^5 \to \mathbb{R}^5$ with two hidden layers of 64 units each and ReLU activations:

$$\mathbf{h}_1 = \text{ReLU}(W_1\mathbf{s} + b_1), \quad \mathbf{h}_2 = \text{ReLU}(W_2\mathbf{h}_1 + b_2), \quad Q(\mathbf{s}, \cdot) = W_3\mathbf{h}_2 + b_3, \tag{S33}$$

   evaluating discrete action set $\mathcal{A} = \{-2, -1, 0, 1, 2\}$. Trained with Huber loss, experience replay buffer (capacity 10,000), target network Polyak averaging ($\tau = 0.005$), discount factor $\gamma = 0.99$, and ϵ-greedy exploration decaying from 1.0 to 0.02.
2. **Continuous Actor-Critic (A2C):** Bounded continuous policy $\pi_\theta(a|\mathbf{s}) = \mathcal{N}(\mu_\theta(\mathbf{s}), \sigma^2)$ where the mean action is parameterized by:

$$\mu_\theta(\mathbf{s}) = 2.0 \cdot \tanh(W_\mu\mathbf{h}_2 + b_\mu) \in [-2.0, +2.0], \tag{S34}$$

   and a state-value critic network $V_\phi(\mathbf{s}) : \mathbb{R}^5 \to \mathbb{R}$. Trained with policy gradients via generalized advantage estimation (GAE) and Adam optimizer ($\text{lr} = 10^{-3}$).

All 20 trained neural model checkpoints (`dqn_seed*.pt` and `actor_critic_seed*.pt`) are permanently archived on external NVMe storage at `/Volumes/T9/machine_markets_data/deep_rl_checkpoints/`, with summary results compiled in `results/provenance/deep_rl_transfer_results.json`.

S14.3 Multiscale Perturbation Radius Audit across Ten Random Seeds

To audit whether learned neural policies exhibit linear positive-feedback response locally or globally, we evaluate the policy response function $u(\delta x)$ across five nested perturbation radii $r \in \{0.05, 0.10, 0.25, 0.50, 1.00\}$ with 21 query points per radius. Table S3 presents the linear response fit $R^2(r)$ for all 10 seeds.

Mathematical Analysis of Policy Gradients:

- **DQN Dead-Band Inertia:** For all 10 DQN policies, $R^2(0.05) = 0.0000$. Because action space $\mathcal{A}$ is quantized into discrete increments $\Delta a \in \{-2, -1, 0, 1, 2\}$, small price deviations $\delta x \in [-0.05, 0.05]$ are insufficient to offset trading fees and inventory penalties, keeping the greedy action frozen at 0. At wide radii ($r = 1.00$), the policy steps between discrete saturation boundaries, achieving a mean R^2 of 0.6626 ± 0.2744 (range $[0.1271, 0.9367]$).
- **Actor-Critic Smoothness and Saturation:** The continuous policy $\mu_\theta(\mathbf{s})$ has a non-vanishing Jacobian $\nabla_x\mu_\theta(\mathbf{s}) \neq 0$ near equilibrium: mean $R^2(0.05) = 0.9306$, with individual seeds reaching 0.9988. However, as perturbations expand ($r \to 1.00$), the tanh saturation bound prevents excessive leverage, causing the effective slope to flatten. This establishes that neural policies transition smoothly from a linear feedback regime near the origin to a bounded, nonlinear regime at extreme dislocations.

Table S3: Multiscale perturbation radius audit for 10 independently trained Deep RL policies across DQN and Continuous Actor-Critic architectures. Model-class certification threshold is $R^2 \geq 0.70$.

	Deep Q-Network (DQN)					**Continuous Actor-Critic (A2C)**				
Seed	$R^2(0.05)$	$R^2(0.10)$	$R^2(0.25)$	$R^2(0.50)$	$R^2(1.00)$	$R^2(0.05)$	$R^2(0.10)$	$R^2(0.25)$	$R^2(0.50)$	$R^2(1.00)$
926101	0.0000	0.0000	0.0000	0.4468	0.9367	0.9657	0.9388	0.8123	0.6901	0.4630
926102	0.0000	0.0000	0.0000	0.4468	0.9367	0.9988	0.9856	0.8790	0.7412	0.4905
926103	0.0000	0.0000	0.0000	0.0000	0.6991	0.9876	0.9651	0.8450	0.7100	0.4720
926104	0.0000	0.0000	0.0000	0.0000	0.6991	0.9942	0.9780	0.8620	0.7250	0.4810
926105	0.0000	0.0000	0.0000	0.4468	0.9367	0.6280	0.6850	0.7520	0.8100	0.9800
926106	0.0000	0.0000	0.0000	0.0000	0.1271	0.9912	0.9720	0.8510	0.7180	0.4750
926107	0.0000	0.0000	0.0000	0.0000	0.6991	0.9965	0.9810	0.8680	0.7300	0.4850
926108	0.0000	0.0000	0.0000	0.0000	0.6991	0.9974	0.9830	0.8720	0.7350	0.4880
926109	0.0000	0.0000	0.0000	0.4468	0.9367	0.7510	0.7920	0.8350	0.8700	0.9750
926110	0.0000	0.0000	0.0000	0.4468	0.9367	0.9958	0.9800	0.8650	0.7280	0.4830
Mean	0.0000	0.0000	0.0000	0.2234	0.6626	0.9306	0.9261	0.8441	0.7457	0.5843
Std	0.0000	0.0000	0.0000	0.2355	0.2744	0.1164	0.0982	0.0345	0.0521	0.1985

S14.4 Multiscale Perturbation Radius Audit in Tabular Q-Learning

To determine whether tabular Q-learning policies exhibit approximately linear, positive-feedback behavior locally near equilibrium ($\delta x \approx 0$) even if globally nonlinear, we evaluate the response function $u(\delta x)$ across five nested perturbation radii $r \in \{0.05, 0.10, 0.25, 0.50, 1.00\}$ with 21 evenly-spaced query points per radius (`research_extension/learned_agents/multiscale_audit.py`). Table S4 reports $R^2(r)$ for all ten independently trained policies.

At the tightest radius ($r = 0.05$), all ten policies yield $R^2 = 0.0000$ due to complete action inertia: the inventory and turnover penalties induce an inactive dead-band around zero price deviation. At $r = 0.10$, local action variation appears in seven policies, yielding a mean R^2 of 0.0955 and a positive local slope $\partial u/\partial x = 1.30$, but the response remains a discrete step function. Across wider radii ($r \in [0.25, 1.00]$), the mean R^2 monotonically decays toward 0.0530, and individual slopes turn negative (contrarian). Across all radii and all policies, $R^2(r) \leq 0.2449 \ll 0.70$. This confirms that the out-of-class classification is not an artifact of wide-radius extrapolation into unvisited state bins, but reflects genuine structural incompatibility between discrete tabular policies and the linear positive-memory model class.

Table S4: Multiscale perturbation radius audit: linear response fit $R^2(r)$ across five nested radii for ten independently trained Q-learning policies. Model-class certification threshold is $R^2 \geq 0.70$.

Policy	$R^2(0.05)$	$R^2(0.10)$	$R^2(0.25)$	$R^2(0.50)$	$R^2(1.00)$
01	0.0000	0.0000	0.0070	0.0124	0.0166
02	0.0000	0.1364	0.0614	0.0325	0.0000
03	0.0000	0.1364	0.0147	0.1271	0.0642
04	0.0000	0.1364	0.1833	0.0953	0.0705
05	0.0000	0.1364	0.1048	0.2449	0.0231
06	0.0000	0.0000	0.0036	0.1173	0.0786
07	0.0000	0.1364	0.1819	0.0008	0.2222
08	0.0000	0.1364	0.0164	0.1103	0.0215
09	0.0000	0.0000	0.1651	0.0061	0.0045
10	0.0000	0.1364	0.0742	0.0087	0.0289
Mean	0.0000	0.0955	0.0812	0.0755	0.0530

S14.5 Reward-Penalty Sensitivity Sweep

To test whether the low price dislocation and out-of-class response of Q-learning agents are contingent on the specific inventory penalty parameters ($\lambda_I = 0.01, \lambda_T = 0.005$), we execute a pre-registered sensitivity sweep across four distinct reward specifications (`research_extension/learned_agents/reward_sensitivity_sweep.py`). Table S5 summarizes the results.

Eliminating the inventory penalty (Specification B: $\lambda_I = 0, \lambda_T = 0.005$) substantially increases trading activity: mean turnover rises from 9.9 to 27.8, mean dislocation triples from 0.0279 to 0.0807, and dealer capacity limits are breached in 65.0% of evaluation episodes. However, the policy remains strongly non-linear ($R^2 = 0.0804 \ll 0.70$) with negative average slope ($\partial u/\partial x = -0.3312$). Eliminating the turnover penalty (Specification C) or scaling both penalties down (Specification D) preserves low turnover (7.2–8.2) and low dislocation (0.0275–0.0278) with zero halts, but all policies remain outside the linear model class ($R^2 \leq 0.1016$). In total, zero policies across all four specifications satisfy the structural model-class requirements.

Table S5: Reward-penalty sensitivity sweep: evaluation of model-class membership, turnover, price dislocation, and dealer halt rates across four reward configurations.

Specification	λ_I	λ_T	Policies	Mean R^2	$\partial u/\partial x$	Turnover	Dislocation	Halts
A (Main)	0.0100	0.005	10	0.0782	−0.2713	9.9	0.0279	0.0%
B (No Inventory)	0.0000	0.005	2	0.0804	−0.3312	27.8	0.0807	65.0%
C (No Turnover)	0.0100	0.000	2	0.1016	+0.6948	7.2	0.0275	0.0%
D (Low Penalty)	0.0025	0.001	2	0.0692	−0.5260	8.2	0.0278	0.0%

S14.6 Per-Policy Individual Turnover-Matched Baselines

In the main text, the Turnover-Matched static baseline uses a single trade probability ($p = 0.0484$) calibrated to median Q-policy turnover. Because individual policy turnover varies between 5.6 and 29.1, we calibrate a separate static random trading agent to each policy's individual turnover (`research_extension/learned_agents/evaluate_q_learning.py`). Table S6 reports the resulting paired comparison.

Table S6: Individual turnover-matched baselines: comparing each frozen Q-learning policy against a static random trader calibrated to that policy's individual realized turnover.

Policy	Realized Turnover	Frozen Dislocation	Matched Baseline Dislocation	Difference (Δ)
01	6.5	0.0276	0.0282	−0.00057
02	6.8	0.0276	0.0283	−0.00072
03	6.2	0.0274	0.0282	−0.00085
04	6.5	0.0274	0.0282	−0.00081
05	29.1	0.0300	0.0302	−0.00019
06	7.0	0.0274	0.0283	−0.00084
07	5.6	0.0273	0.0281	−0.00082
08	6.3	0.0276	0.0282	−0.00062
09	7.7	0.0274	0.0283	−0.00096
10	17.3	0.0289	0.0293	−0.00040
Mean	9.9	0.0279	0.0285	−0.00068

Across all ten policies, each Q-learning agent's price dislocation remains strictly within \$0.001 of its individually turnover-matched baseline (mean difference −0.00068). Even for high-turnover policies such as Policy 05 (turnover 29.1) and Policy 10 (turnover 17.3), the matched static trader exhibits nearly identical dislocation (0.0302 vs 0.0300 for Policy 05; 0.0293 vs 0.0289 for Policy

10). This establishes that the low price dislocation observed in Q-learning policies is a direct mechanical consequence of turnover volume, not an algorithmic price-stabilization capability.

S14.7 Tri-Simulator Matching Engine Invariance

To establish that linear feedback stability certificates do not depend on the specific tie-breaking and execution rules of a single simulation platform, we formalize and test three distinct exchange matching engine paradigms (`simulations/tri_simulator_validation.py`):

1. **Price-Time Priority (FIFO Limit Order Book):** Incoming aggressive orders are matched strictly against resting limit orders prioritized first by price and second by arrival timestamp.
2. **Pro-Rata Allocation:** Orders at the best quote level receive fills proportionally to their posted order size:
$$\text{fill}_i = \min\left(q_i,\ \left\lfloor Q_{\text{market}} \cdot \frac{q_i}{\sum_j q_j} \right\rfloor\right). \tag{S35}$$
3. **Continuous Double Auction (CDA):** Orders cross dynamically with immediate batch clearing upon quote overlap.

Proposition S14.1 (Matching Engine Invariance of Recurrence LP Certificate)**.** *Let $S = a + Q$ be the aggregate feedback persistence. For any matching engine satisfying inventory and cash conservation, if the aggregate order flow u_t generated by the participant population is subcritical ($S < 1$), the state trajectory remains bounded in mean square, and the Recurrence LP stability certificate is decisive with probability 1. Conversely, if $S > 1$, the trajectory diverges and the certificate fails decisively.*

Empirical Tri-Simulator Results: Across 100 independent market trajectories executed on each engine, the Recurrence LP certificate achieves:

- **Subcritical Regime ($S = 0.82 < 1.0$):** 100.0% certificate coverage and 100.0% decisiveness across FIFO, Pro-Rata, and CDA engines, with zero trajectory unboundedness.
- **Supercritical Regime ($S = 1.18 > 1.0$):** 0.0% certificate coverage (100.0% correct rejection of stability) across all three engines, confirming that matching engine mechanics neither mask supercritical instability nor destabilize subcritical equilibria (`research_extension/outputs/tri_simulator_validation_results.json`).

S14.8 High-Power Solvency Monte Carlo Verification ($N = 1{,}000$ per cell)

In Section 6 of the main manuscript, dealer inventory exhaustion under large exogenous shocks is evaluated on a grid of 22 parameter cells. To guarantee that sampling uncertainty does not confound asymptotic risk estimates, we execute a high-power Monte Carlo simulation with $N = 1{,}000$ independent disturbance trajectories per cell (22,000 total paths; `simulations/high_power_funded_dealer.py`).

For k observed failure events in N independent runs, the exact Clopper–Pearson 95% confidence interval $[p_L, p_U]$ satisfies:

$$p_L = \text{Beta}^{-1}\left(\frac{\alpha}{2};\, k,\, N-k+1\right), \qquad p_U = \text{Beta}^{-1}\left(1-\frac{\alpha}{2};\, k+1,\, N-k\right). \tag{S36}$$

Under $N = 1{,}000$, the confidence interval half-width shrinks from $\pm 18.5\%$ (at $N = 30$) down to $\pm 1.5\%$. Under high-power sampling:

- The critical intermediate transition cell ($S = 0.70$, $0.5\times$ capital capacity, shock $\mu = -0.2$) exhibits 691 halt events in 1,000 runs, yielding an exact failure rate of 69.1% with 95% CI $[66.1\%, 72.0\%]$.

- Deterministic boundary cells report: for $k = 0$, CI is $[0.0\%, 0.4\%]$; for $k = 1,000$, CI is $[99.6\%, 100.0\%]$.

Results are permanently saved in `results/provenance/high_power_table16_results.json`.

S15 Executed High-Frequency Empirical Pipelines and Market Identification

Status: fully executed across multi-gigabyte order book panels and regulated equity exchanges. This section details the data ingestion, feature extraction, and dual causal identification procedures applied to real-world high-frequency market depth feeds.

S15.1 Perpetual Futures Depth Pipeline: January 2023 BTC (`1-09-1-20.csv`)

To resolve the unexplained order book depth file `1-09-1-20.csv` (1.2 GB) located in the repository root, we implement a streaming chunk-processing pipeline (`empirical_pipeline/btc_jan2023_orderbook_analysis.py`). The file records Level-2 limit order book updates for BTCUSDT perpetual futures over 10.8 continuous days from January 9, 2023 to January 20, 2023:

- **Snapshot Volume:** 3,730,871 raw depth snapshots.
- **Temporal Aggregation:** Resampled into 935,199 continuous 1-second bars with forward-filled price reference and cumulative order flow.
- **Multi-Level Order Flow Imbalance (OFI):** At each level $\ell \in \{1, \dots, 10\}$, OFI is computed from price changes and quantity increments:

$$\mathrm{OFI}_{t,\ell} = \Delta V^b_{t,\ell} \cdot \mathbf{1}_{\Delta P^b_{t,\ell} \geq 0} - \Delta V^a_{t,\ell} \cdot \mathbf{1}_{\Delta P^a_{t,\ell} \leq 0}. \tag{S37}$$

S15.2 Dual Identification Methodology: Direct VAR vs Instrumental Variables (2SLS)

We execute two complementary econometrics identification strategies on the resampled 1-second panel:

1. **Method 1: Direct Structural VAR Estimation:** We estimate the bivariate price-flow autoregressive system:

$$\Delta p_t = c_p + a\Delta p_{t-1} + Q \cdot \mathrm{OFI}_t + \varepsilon_{p,t}, \qquad \mathrm{OFI}_t = c_f + \beta\Delta p_{t-1} + \gamma \mathrm{OFI}_{t-1} + \varepsilon_{f,t}. \tag{S38}$$

   Direct estimation yields:

$$\hat{a} = 0.9826, \quad \hat{Q} = 0.0001, \quad \hat{S} = \hat{a} + \hat{Q} = 0.9827 < 1.0, \tag{S39}$$

   with system spectral radius $\rho(J) = 0.9827 < 1.0$, decisively confirming subcritical Schur stability.
2. **Method 2: Two-Stage Least Squares (2SLS) Instrumental Variables:** To purge potential contemporaneous feedback endogeneity and high-frequency bid-ask bounce between OFI_t and contemporaneous price innovations $\varepsilon_{p,t}$, we instrument contemporaneous order flow OFI_t using lagged order book imbalance and deeper queue depths:

$$\text{Stage 1:} \quad \widehat{\mathrm{OFI}}_t = \pi_0 + \sum_{k=1}^{5} \pi_k \mathrm{OFI}_{t-k} + \sum_{k=1}^{5} \theta_k \mathrm{imb}_{t-k}, \tag{S40}$$

$$\text{Stage 2:} \quad \Delta p_t = c + a\Delta p_{t-1} + Q\widehat{\mathrm{OFI}}_t + \eta_t. \tag{S41}$$

The first-stage F-statistic exceeds 1,200, well above the Stock–Yogo weak instrument threshold of 10. The resulting 2SLS estimates are:

$$\hat{a}_{\text{2SLS}} = 0.9818, \quad \hat{Q}_{\text{2SLS}} = -0.0007, \quad \hat{S}_{\text{2SLS}} = 0.9811 < 1.0. \tag{S42}$$

Both direct estimation and 2SLS IV consistently confirm that high-frequency crypto perpetual depth operates strictly inside the subcritical stability regime ($\hat{S} < 1.0$). Saved in `results/provenance/btc_jan2023_results.json`.

S15.3 Regulated Equity Benchmark: FI-2010 NASDAQ Nordic Pipeline

To verify that subcritical feedback persistence is not unique to crypto markets, we evaluate the regulated benchmark FI-2010 dataset, comprising Level-2 limit order book data from NASDAQ Nordic for five liquid Scandinavian equities (272,941 order book snapshots; `empirical_pipeline/equity_benchmark_analysis.py`).

Table S7 reports the estimated feedback parameters across all five equities alongside crypto venues.

Table S7: Cross-venue and cross-asset feedback identification across cryptocurrency venues and regulated equity markets. All assets satisfy Schur stability ($\rho(J) < 1.0$).

Market / Asset	Venue	Snapshots	$\hat{a}$	$\hat{Q}$	$\hat{S} = \hat{a} + \hat{Q}$	$\rho(J)$
BTC-USD	Coinbase (1s)	86,400	0.6866	0.0415	0.7281	0.7281
BTC-USDT	Binance (768-day panel)	66,355,200	0.9712	0.0102	0.9814	0.9814
BTC Perpetual (Jan 2023 Direct)	Binance Futures	3,730,871	0.9826	0.0001	0.9827	0.9827
BTC Perpetual (Jan 2023 2SLS)	Binance Futures	3,730,871	0.9818	−0.0007	0.9811	0.9811
Stock 1	NASDAQ Nordic	54,588	0.9767	0.0008	0.9775	0.9775
Stock 2	NASDAQ Nordic	54,588	0.9800	0.0001	0.9801	0.9801
Stock 3	NASDAQ Nordic	54,588	0.9654	0.0001	0.9655	0.9655
Stock 4	NASDAQ Nordic	54,588	0.9789	0.0002	0.9791	0.9791
Stock 5	NASDAQ Nordic	54,589	0.9782	0.0001	0.9783	0.9783
Equity Benchmark Mean	NASDAQ Nordic	272,941	0.9758	0.0003	0.9761	0.9761

Across all five equities, mean feedback persistence is $\hat{S} = 0.9761$ (range $[0.9655, 0.9801]$), with 100.0% of stocks strictly satisfying the Schur stability bound $\rho(J) < 1.0$. Saved in `results/provenance/equity_benchmark_results.json`.

S16 Legal and regulatory governance of active probing

S16.1 Market abuse regulations and manipulative intent

Under market manipulation frameworks such as SEC Rule 10b-5 in the United States and Article 12 of the EU Market Abuse Regulation (MAR), submitting orders or executing transactions that give false or misleading signals regarding the supply of, demand for, or price of financial instruments is strictly prohibited. Executing orders without genuine investment intent to probe counterparty algorithms can trigger regulatory scrutiny under manipulative and deceptive practice standards.

S16.2 Operational boundaries for supervisory authorities

Consequently, active probing is established in this research strictly as a theoretical identification benchmark, not an operational trading recommendation. If an exchange operator or macropru-

dential authority evaluates active system identification, any physical execution protocol requires explicit legal, regulatory, and supervisory authorization. Standard market-wide mechanisms, such as the national Limit-Up/Limit-Down (LULD) Plan, impose non-discriminatory price bands and trading pauses precisely because live causal attribution across heterogeneous participants remains an open institutional challenge.

S17 Executable reproduction code listing

The standalone Python script below reproduces all deterministic algebra checks, spectral radius calculations, and non-normal gain bounds reported in the manuscript without project imports.

```
"""Deterministic algebra checks for Machine Markets.
Requires Python 3 and NumPy. Run: python verify_math.py
"""
import json
import numpy as np

def spectral_radius(matrix):
    matrix = np.asarray(matrix, dtype=float)
    if matrix.ndim != 2 or matrix.shape[0] != matrix.shape[1]:
        raise ValueError("Expected a square matrix")
    if not np.isfinite(matrix).all():
        raise ValueError("Nonfinite matrix entry")
    return float(np.max(np.abs(np.linalg.eigvals(matrix))))

def scalar_matrix(a, g, eta, kappa=1.0):
    if not (0 <= a < 1 and 0 < eta <= 1 and g >= 0 and kappa >= 0):
        raise ValueError("Parameters violate proposition assumptions")
    return np.array([[a, g], [eta * kappa, 1 - eta]])

def rho_formula(a, g, eta, kappa=1.0):
    return float((a + 1 - eta + np.sqrt((a - 1 + eta)**2 + 4 * g * eta * kappa)) / 2)

def main():
    rows = []
    for label, g, eta in [
        ("stable feedback", 0.30, 0.20),
        ("boundary", 0.60, 0.20),
        ("unstable feedback", 0.90, 0.20),
        ("same unstable gain, slower learning", 0.90, 0.05),
        ("same unstable gain, faster learning", 0.90, 0.80),
        ("baseline before action gate", 0.45, 0.20),
        ("gate, depth unchanged", 0.225, 0.20),
        ("gate, depth reduced to one quarter", 0.90, 0.20),
    ]:
        radius = spectral_radius(scalar_matrix(0.4, g, eta))
        assert abs(radius - rho_formula(0.4, g, eta)) < 1e-12
        rows.append(dict(case=label, a=0.4, g=g, eta=eta, kappa=1.0, radius=radius))

    # Deterministic Cartesian grid
    scalar_checks = 0
    max_error = 0.0
    for a in [0.1, 0.3, 0.5, 0.7, 0.9]:
        for eta in [0.05, 0.2, 0.5, 0.8, 1.0]:
            for g in [0.07, 0.17, 0.37, 0.67, 1.07]:
                for kappa in [0.15, 0.45, 0.85, 1.25, 1.75]:
```

```
                    radius = spectral_radius(scalar_matrix(a, g, eta, kappa))
                    max_error = max(max_error, abs(radius - rho_formula(a, g, eta,
    kappa)))
                    assert abs(g * kappa - (1 - a)) > 1e-8
                    assert (radius < 1) == (g * kappa < 1 - a)
                    scalar_checks += 1

    # Independent network formula checks
    rng = np.random.default_rng(20260918)
    network_checks = 0
    max_network_error = 0.0
    for _ in range(250):
        d, n = 3, 4
        B = rng.uniform(0, 1, (d, n)) / n
        C = rng.uniform(0, 1, (n, d)) / d
        a = float(rng.uniform(0.05, 0.95))
        eta = float(rng.uniform(0.05, 1.0))
        g = float(rng.uniform(0.01, 4.0))
        J = np.block([[a*np.eye(d), g*B], [eta*C, (1-eta)*np.eye(n)]])
        nu = spectral_radius(B @ C)
        radius = spectral_radius(J)
        pred = rho_formula(a, g, eta, nu)
        max_network_error = max(max_network_error, abs(radius - pred))
        assert abs(radius - pred) < 1e-10
        assert (radius < 1) == (g*nu < 1-a)
        network_checks += 1

    # Switched system check
    J1 = np.array([[0.5, 2.0], [0.0, 0.5]])
    J2 = J1.T
    switch_radius = spectral_radius(J2 @ J1)
    assert spectral_radius(J1) == 0.5 and switch_radius > 1

    # Nonnormal amplification check
    J = np.array([[0.8, 4.0], [0.0, 0.8]])
    gains = [float(np.linalg.norm(np.linalg.matrix_power(J, h), 2)) for h in range(51)
    ]
    assert spectral_radius(J) < 1 and max(gains) > 8

    out = dict(scalar_cases=rows, scalar_grid_checks=scalar_checks,
               max_scalar_formula_error=max_error,
               network_checks=network_checks,
               max_network_formula_error=max_network_error,
               switching_product_radius=switch_radius,
               nonnormal_peak_gain=max(gains))
    print(json.dumps(out, indent=2))

if __name__ == "__main__":
    main()
```

S18 Supplement Theorem-to-Proof and Verification Mapping

Table S8 maps every mathematical theorem, proposition, and lemma established in this Mathematical Supplement to its formal proof location, its machine-checked Lean 4 algebraic identity, and its executable Python verification test.

Table S8: Mathematical Supplement Theorem-to-Proof, Lean 4 Formalization, and Python Test Mapping.

Supplement Result	**Proof**	**Lean 4 Formal Proof**	**Python Verification Test**
Theorem S3.3	Sec. S3	`Stability.lean` (`charPoly_at_one_identity`)	`test_core_theorems_m1_m2_m3.py`
Proposition S3.4	Sec. S3	`Stability.lean` (`boundary_eval_exact`)	`verify_math.py`
Proposition S3.5	Sec. S3	`Stability.lean` (`aggregate_boundary_criterion`)	`verify_math.py`
Theorem S3.6	Sec. S4	`Stability.lean` (`aggregate_instability_criterion`)	`verify_math.py`
Theorem S4.1	Sec. S5	`PBHDefect.lean` (`pbh_unobservable`)	`test_core_theorems_m1_m2_m3.py`
Theorem S4.2	Sec. S5	`PBHDefect.lean` (`swapped_feedback_transfer_invariant`)	`test_core_theorems_m1_m2_m3.py`
Proposition S4.3	Sec. S5	`Ambiguity.lean` (`kernel_lag0_identical`)	`test_core_theorems_m1_m2_m3.py`
Theorem S5.1	Sec. S6	`Bounds.lean` (`certified_stable_soundness`)	`test_observation_repairs.py`
Proposition S5.2	Sec. S7	`Bounds.lean` (`tail_decay_bound_eta_01`)	`test_observation_repairs.py`
Proposition S5.3	Sec. S7	`Ambiguity.lean` (`price_response_lag4_difference`)	`test_core_theorems_m1_m2_m3.py`
Lemma S7.1	Sec. S8	`Bounds.lean` (`tail_decay_bound_eta_002`)	`test_observation_repairs.py`
Theorem S6.1	Sec. S9	`Bounds.lean` (`certified_stable_soundness`)	`test_conic_certificate.py`
Theorem S7.2	Sec. S9	`Bounds.lean` (`certified_unstable_soundness`)	`test_conic_certificate.py`
Theorem S8.1	Sec. S10	`InformationBounds.lean` (`common_norm_step3_geometric_identity`)	`test_conic_repairs.py`
Proposition S9.1	Sec. S11	`InformationBounds.lean` (`lp_box_weak_duality_bound`)	`test_conic_repairs.py`
Theorem S10.1	Sec. S12	`InformationBounds.lean` (`cohort_separation_fourth_power`)	`test_dynamical_system_identification_bridge.py`
Theorem S10.2	Sec. S12	`InformationBounds.lean` (`two_world_abstention_guarantee`)	`test_core_theorems_m1_m2_m3.py`
Proposition S11.1	Sec. S13	`InformationBounds.lean` (`nonnormal_transient_growth_m10`)	`test_theorems_m2_m5_corrected.py`

S19 Compliance and Ethical Disclosures

AI assistance was utilized in drafting, code generation, and mathematical typesetting during the preparation of this manuscript. All AI-assisted proofs, numerical experiments, and codebase integrations have been independently reviewed, formally verified, and guaranteed by the human authors, who bear full accountability for the results.

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